CLab 4

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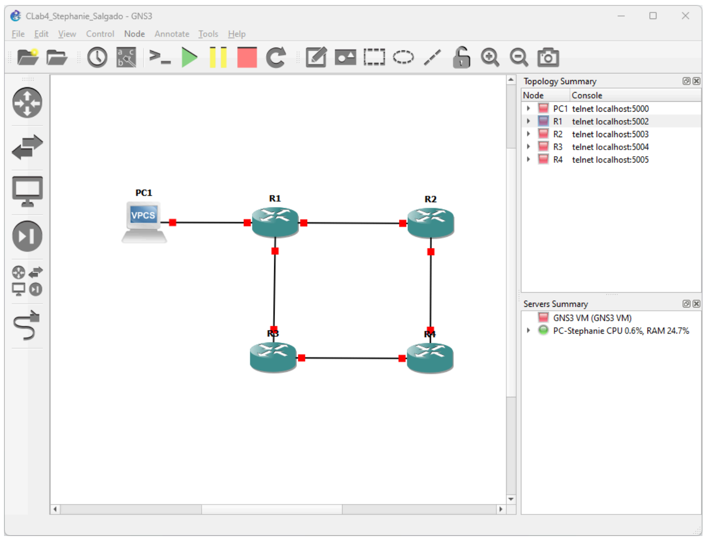
Creating the network



I created the network with 1 VPC and 4 routers by dragging them into the workspace and adding a connection between them.

Console connecting to all nodes

PC1

```
PC1> ip 192.168.1.2 255.255.255.0 gateway 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0 gateway 192.168.1.1

PC1> save
Saving startup configuration to startup.vpc
.  done
```

R1

```
R1#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
*Sep 26 11:13:55.271: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Sep 26 11:13:56.271: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#interface f1/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
*Sep 26 11:15:07.755: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Sep 26 11:15:08.755: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config-if)#exit
R1(config)#interface f2/0
R1(config-if)#ip address 192.168.13.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
*Sep 26 11:15:59.495: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Sep 26 11:16:00.495: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R1(config-if)#exit
```

R2

```
R2#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface f0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#
*Sep 26 12:03:02.471: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Sep 26 12:03:03.471: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#interface f1/0
R2(config-if)#ip address 192.168.24.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#
*Sep 26 12:03:43.399: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Sep 26 12:03:44.399: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R2(config-if)#exit
```

R3

```
R3#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface f0/0
R3(config-if)#ip address 192.168.13.2 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#
*Sep 26 11:22:57.479: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Sep 26 11:22:58.479: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config-if)#exit
R3(config)#interface f1/0
R3(config-if)#ip address 192.168.34.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#
*Sep 26 11:25:03.671: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Sep 26 11:25:04.671: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R3(config-if)#exit
```

R4

```
R4#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#interface f0/0
R4(config-if)#ip address 192.168.24.2 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#
*Sep 26 11:26:32.767: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Sep 26 11:26:33.767: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config-if)#exit
R4(config)#interface f1/0
R4(config-if)#ip address 192.168.34.2 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#
*Sep 26 11:27:09.783: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Sep 26 11:27:10.783: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R4(config-if)#exit
```

Configuring loopback with R4:

```
R4(config)#interface loopback0
R4(config-if)#
*Sep 26 11:30:24.383: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R4(config-if)#ip address 4.4.4.4 255.255.255.255
R4(config-if)#exit
```

Notice all routers now have a basic configuration. Only R4 has loopback configured.

Configuring OSPF:

R1

```
R1(config)#router ospf 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.12.0 0.0.0.255 area 0
R1(config-router)#network 192.168.13.0 0.0.0.255 area 0
R1(config-router)#exit
R1(config)#
```

R2

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 0
R2(config-router)#network 192.168.24.0 0.0.0.255 area 0
R2(config-router)#exit
R2(config)#
```

R3

```
R3(config)#router ospf 1
R3(config-router)#network 192.168.13.0 0.0.0.255 area 0
R3(config-router)#network 192.168.34.0 0.0.0.255 area 0
R3(config-router)#exit
R3(config)#
```

R4

```
R4(config)#router ospf 1
R4(config-router)#network 4.4.4.4 0.0.0.0 area 0
R4(config-router)#network 192.168.24.0 0.0.0.255 area 0
R4(config-router)#network 192.168.34.0 0.0.0.255 area 0
R4(config-router)#exit
R4(config)#
```

Now, all routers have OSPF configured.

Show ip route on all routers:

R1

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/3] via 192.168.12.2, 00:00:25, FastEthernet1/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, FastEthernet0/0
L        192.168.1.1/32 is directly connected, FastEthernet0/0
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.12.0/24 is directly connected, FastEthernet1/0
L        192.168.12.1/32 is directly connected, FastEthernet1/0
      192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.13.0/24 is directly connected, FastEthernet2/0
L        192.168.13.1/32 is directly connected, FastEthernet2/0
O     192.168.24.0/24 [110/2] via 192.168.12.2, 00:02:38, FastEthernet1/0
O     192.168.34.0/24 [110/3] via 192.168.12.2, 00:00:15, FastEthernet1/0
R1#
```

R2

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/2] via 192.168.24.2, 00:04:42, FastEthernet1/0
O     192.168.1.0/24 [110/2] via 192.168.12.1, 00:07:03, FastEthernet0/0
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.12.0/24 is directly connected, FastEthernet0/0
L        192.168.12.2/32 is directly connected, FastEthernet0/0
O     192.168.13.0/24 [110/2] via 192.168.12.1, 00:07:03, FastEthernet0/0
      192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.24.0/24 is directly connected, FastEthernet1/0
L        192.168.24.1/32 is directly connected, FastEthernet1/0
O     192.168.34.0/24 [110/2] via 192.168.24.2, 00:04:22, FastEthernet1/0
R2#
```

R3

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/2] via 192.168.34.2, 00:04:37, FastEthernet1/0
O     192.168.1.0/24 [110/4] via 192.168.34.2, 00:04:37, FastEthernet1/0
O     192.168.12.0/24 [110/3] via 192.168.34.2, 00:04:37, FastEthernet1/0
      192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.13.0/24 is directly connected, FastEthernet0/0
L        192.168.13.2/32 is directly connected, FastEthernet0/0
O     192.168.24.0/24 [110/2] via 192.168.34.2, 00:04:37, FastEthernet1/0
      192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.34.0/24 is directly connected, FastEthernet1/0
L        192.168.34.1/32 is directly connected, FastEthernet1/0
R3#
```

R4

```
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      4.0.0.0/32 is subnetted, 1 subnets
C        4.4.4.4 is directly connected, Loopback0
O     192.168.1.0/24 [110/3] via 192.168.24.1, 00:05:00, FastEthernet0/0
O     192.168.12.0/24 [110/2] via 192.168.24.1, 00:05:00, FastEthernet0/0
O     192.168.13.0/24 [110/2] via 192.168.34.1, 00:04:40, FastEthernet1/0
      192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.24.0/24 is directly connected, FastEthernet0/0
L        192.168.24.2/32 is directly connected, FastEthernet0/0
      192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.34.0/24 is directly connected, FastEthernet1/0
L        192.168.34.2/32 is directly connected, FastEthernet1/0
R4#
```

Notice the "C" (connected), "L" (local), and "O" (OSPF) codes and the subnet 4.4.4.4 (loopback configured in R4). There should now be communication across all routers. Some will be "C" and"L", but the rest will be "O" codes.

Pinging loopback:

```
PC1> ping 4.4.4.4
84 bytes from 4.4.4.4 icmp_seq=1 ttl=253 time=75.296 ms
84 bytes from 4.4.4.4 icmp_seq=2 ttl=253 time=75.760 ms
84 bytes from 4.4.4.4 icmp_seq=3 ttl=253 time=75.788 ms
84 bytes from 4.4.4.4 icmp_seq=4 ttl=253 time=75.742 ms
84 bytes from 4.4.4.4 icmp_seq=5 ttl=253 time=75.393 ms
```

Directing ICMP traffic from PC1 through R1 and R3 link:

```
R1#show route-map
R1#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip access-list extended ICMP_H1
R1(config-ext-nacl)#permit icmp host 192.168.1.100 host 4.4.4.4
R1(config-ext-nacl)#exit
R1(config)#route-map PBR_H1 permit 10
R1(config-route-map)#match ip address ICMP_H1
R1(config-route-map)#set ip next-hop 192.168.13.2
R1(config-route-map)#exit
R1(config)#ip local policy route-map PBR_H1
R1(config)#exit
```

To do this, I configured R1. Notice at first using "show route-map" had no output. The configuration should allow traffic from R1 to 4.4.4.4 to be redirected toward R3. The "ip local policy route-map PBR_H1" command should enable this.

```
R1#show route-map
route-map PBR_H1, permit, sequence 10
  Match clauses:
    ip address (access-lists): ICMP_H1
  Set clauses:
    ip next-hop 192.168.13.2
  Policy routing matches: 0 packets, 0 bytes
R1#
```

This time, using the command displayed the route map configured above.

```
R1#ping 4.4.4.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:

*Sep 26 13:19:41.583: IP: s=192.168.12.1 (local), d=4.4.4.4, len 100, policy match
*Sep 26 13:19:41.583: IP: route map PBR_R1, item 10, permit
*Sep 26 13:19:41.583: IP: s=192.168.12.1 (local), d=4.4.4.4 (FastEthernet2/0), len 100, policy routed
*Sep 26 13:19:41.583: IP: local to FastEthernet2/0 192.168.13.2.
*Sep 26 13:19:43.583: IP: s=192.168.12.1 (local), d=4.4.4.4, len 100, policy match
*Sep 26 13:19:43.587: IP: route map PBR_R1, item 10, permit
```

```
R1#trace 4.4.4.4
Type escape sequence to abort.
Tracing the route to 4.4.4.4
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.12.2 28 msec 28 msec 28 msec
  2 192.168.24.2 64 msec 60 msec 56 msec
R1#
*Sep 26 13:21:50.683: IP: s=192.168.12.1 (local), d=4.4.4.4, len 28, policy rejected -- normal forwarding
*Sep 26 13:21:50.711: IP: s=192.168.12.1 (local), d=4.4.4.4, len 28, policy rejected -- normal forwarding
*Sep 26 13:21:50.743: IP: s=192.168.12.1 (local), d=4.4.4.4, len 28, policy rejected -- normal forwarding
```

Summary

For this lab, I simulated a network with 4 routers and 1 VPCs in order to try demonstrating policy based routing. I used a basic configuration for all routers and then configured OSPF on them. The goal was to redirect ICMP traffic passing through R1 to 4.4.4.4 through R3. I was unable to prove it was working. However, in theory, the route map displayed should correspond to the correct routing policy to make this happen. Overall, this lab made good use of OSPF like the previous lab, but I didn't seem to get the desired results when I tried pinging 4.4.4.4 from R1 or when I performed a "trace 4.4.4.4".